

Research papers

Rainfall estimation using measurement report data from time-division long term evolution networks

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ABSTRACT

Accurate rainfall measurement is essential for flood forecasting, effective water resources management, agriculture, etc. Many traditional rainfall monitoring methods, such as rainfall gauges, weather radars, and satellites, have been developed. Recently, a rainfall retrieval algorithm based on the radio connection between one telephone tower and another has seen widespread use across countries as it does not require any deployment procedures or costs. However, because underground fibre-optic cable networks have been widely introduced and deployed between telephone towers, usage of rainfall retrieval algorithms has reduced. Measurement report (MR) data from time-division long term evolution (TD-LTE) networks for providing instructions to the user terminal has the potential to estimate rainfall owing to the impact of rainfall-based attenuation on MR data. We present an analysis of MR data collected during the local flood season over 2 months (May and August 2019) in Huizhou city, southern China. Comparisons between estimated and observed rainfall exhibit promising results. Rainfall estimation from MR data is a potential alternative/complementary rainfall estimation method, as well as having use in runoff simulations, with acceptable accuracy (>0.8); this is particularly relevant in the context of disappearing microwave links between telephone towers in cellular communication networks. The rainfall estimation method proposed herein can also be implemented in rarely gauged or ungauged areas for flood simulation.

1. Introduction

Rainfall is the main source of water for the hydrological system and, thus, accurate rainfall monitoring is of great importance for effective water resource management, flood forecasting, agriculture, and many other applications (Gosset et al., 2016). However, the coverage and accuracy of traditional tools (e.g., gauges, weather radars, and satellites) is limited for various technical reasons (Messer and Sendik, 2015). High coverage by gauges involves high costs and most of the earth's surface lacks ground-based rain gauges and weather radars (Heistermann et al., 2013; Tang et al., 2016); the number of reporting rain gauges in Europe, South America, and Africa are declining dramatically (Lorenz and Kunstmann, 2012). Despite the increased coverage and spatio-temporal resolution of satellite rainfall products, from TRMM to GPM, measurement errors and sampling uncertainties limit their applicability (Barrett and Beaumont, 1993; Kidd and Huffman, 2011; Hou et al., 2014; Tang

et al., 2016). Cellular communication networks cover a large fraction of the earth's surface (via <http://opensignal.com/networks>) and, with increasing network densities in proportion to the growing population (Gosset et al., 2016), they have seen substantial use as an alternative source of rainfall information (Messer et al., 2006; Leijnse et al., 2007).

The idea of using cellular communication networks for rainfall estimation is based on microwave attenuation. Rainfall can attenuate the electromagnetic signals transmitted from the antenna of one telephone tower to another. The reduction in the received signal level with respect to the reference signal (measured under dry weather conditions) can be attributed to rainfall-induced-path-integrated attenuation. The relationship between the attenuation and rain rate (i.e., the path-averaged rainfall intensity) has been theoretical justified as a power law (Olsen et al., 1978). To monitor the quality of communication networks, received signal level strength is often stored by communication companies. Rainfall estimation from commercial cellular

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telecommunication networks has become a promising method of rainfall measurement around the world due to the large network coverage and high density of microwave links without additional cost (Messer et al., 2006; Gosset et al., 2016; Messer, 2007). Many relevant techniques and algorithms have been collected (e.g., the microwave tomographic inversion technique (MTIT), Giuli, et al, 1991; Giuli et al., 1999). One of the favourable algorithms is the operational (real-time) rainfall retrieval algorithm; namely, the package “RAINLINK”, developed by Overeem et al. (2016a). “RAINLINK” has also seen application in the Netherlands for real-time rainfall monitoring (Overeem et al., 2013) and is publicly available (via <https://github.com/overeem11/RAINLINK>). Most existing methods or algorithms are driven by microwave link data between telephone towers. Since the wide-spread introduction of underground fibre-optic cable networks for telecommunication between telephone towers (Overeem et al., 2016a, 2016b), especially in China, the majority of link data have become unavailable from commercial cellular communication companies. Therefore, the algorithms for estimating rainfall from this type of microwave link data between telephone towers present too many challenges or are no longer feasible.

The connection between the user terminal and the serving base station (i.e., telephone tower) during communication is developed as a time-division long term evolution (TD-LTE) network and is not replaced by underground fibre-optic cables. To reflect the coverage and interference of the TD-LTE network, measurement report (MR) data has been widely used in practical network evaluation and optimisation for instructing the user terminal to trigger events, such as handover (Chowdhury et al., 2011; He et al., 2013; MIIT, (The Ministry of Industry and Information Technology of the People's Republic of China), 2015; Zhou and Zhao, 2018). According to the regulations of an operator in China (i.e., China mobile), the cycle of MR data collection is controlled by the operation and maintenance centre-radio every 15 min (MIIT, (The Ministry of Industry and Information Technology of the People's Republic of China), 2015). The key fields in the MR data file refer to the reference signal receiving quality power (RSRP), which is the absolute signal strength of the LTE reference signal in dBm (i.e., absolute signal strength compared to a reference level of 1 mW). As the RSRP consists of the reference signal transmission power, antenna gain of the base station, and propagation loss and antenna gain of the terminal, it is used to measure the coverage of the LTE cell on the downlink (i.e., from the base

transceiver station to user equipment). If the RSRP measurement results are categorized into different ranges, the proportions of each range can be determined by the quotient of the number of samples within the threshold range and the total number of samples. The proportion of weak coverage samples (e.g., with RSRP < -110 dBm) can be used to evaluate the performance of the TD-LTE network. Changes to the proportions can be attributed to changing propagation losses, including the impact on microwave attenuation due to rainfall. Then, it is possible to establish a relationship between the rainfall rate and proportions of RSRP from the MR data, which is the focus of our study.

This paper is organized as follows. First, descriptions of the rain gauge, MR, runoff, and pan evaporation data are given in a case study. Next, the methodology for deriving rainfall rate from the MR data, the interpolation for rain gauge data, and the hydrological model for simulating runoff are described. Subsequently, the verification results — illustrating the effectiveness of the rainfall retrieval algorithm and its limitations and applicability to other regions — are presented. Finally, conclusions are provided.

2. Study area and Data

2.1. Study area and gauges data

Andunshui River basin, a tributary of the Dongjiang River basin, is located in the rural area of Huidong county, Huizhou city, China (Figure 1), and has an area of 385 km².

There are six automatic rain gauge stations, each with 30 min rainfall data series over two months (May and August 2019). The 30 min streamflow data for these two months were obtained from the Jiuzhou gauge station, which is located at the outlet of the Andunshui River basin. The mean areal rainfall was estimated by the Thiessen polygon method, based on the six automatic rain gauge stations, and was then taken as the rainfall in this study. The potential evaporation with 30 min time steps was measured at the Jiuzhou station. All rainfall, evaporation, and runoff data were provided by the Bureau of Hydrology, Huizhou city, Guangdong province.

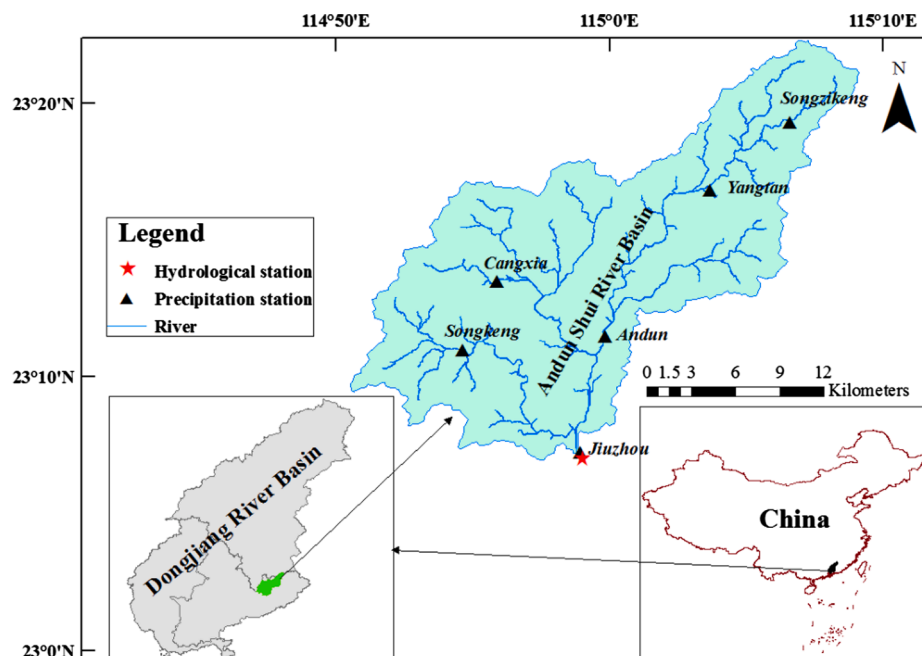


Figure 1. Locations of the gauge rainfall and runoff stations in the study area

2.2. MR Data from TD-LTE network

The MR data is recorded in XML files and from the TD-LTE MR data acquisition system. From the MR data, the reporting range of RSRP is defined from -120 to -60 dBm, with a resolution of 1 dB or 2 dB. The proportions of each range are often analysed to determine the TD-LTE network coverage via commercial cellular communication companies; in this study, this will be used to estimate rainfall. There are four LTE cells on the downlink in the study area monitored by China Mobile Communications Group Guangdong, Huizhou branch. The proportions of RSRP are obtained from these four LTE cells. Although the MR data is recorded every 15 min, it was converted into 30 min data series via the arithmetic mean function of the two records during the 30 min from the available rainfall and runoff gauge data. The changes in RSRP can indicate path-averaged rainfall (horizontal) while the path delay of global positioning system (GPS) signals can help evaluate precipitable water vapour (vertical) (Liou et al., 2001; Yeh et al., 2016). Thus, the GPS signals data from four telephone towers in the study area have also been collected to improve rainfall estimation.

3. Methodology

The process of determining rainfall from the MR data starts with wet (rainy)-dry (non-rainy) classification. Once the rainy and non-rainy time intervals have been classified, the relationship between the amount of areal rainfall and the proportion of the RSRP ranges from the MR data during the raining time interval can be determined for rainfall estimation. After rainfall estimation from the MR data through the determined relationship, a hydrological model is employed to prove the applicability of estimating rainfall from the MR data during runoff simulations. The proportions of RSRP from the TD-LTE network are often correlated with each other and, generally, lower the efficiency for wet-dry classification and rainfall estimation; hence, the principal variables from the proportions of RSRP require determining. Principal component analysis (PCA) is used as a dimensionality reduction method for selecting the principal variables. The entire process of the work presented herein is shown in Figure 2.

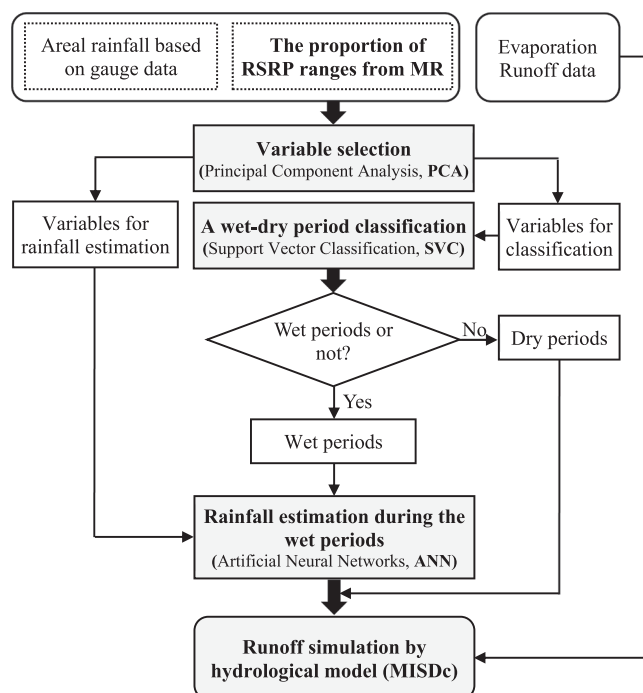


Figure 2. Flowchart of rainfall estimation from MR data

3.1. The principal component analysis (PCA) method for variable selection

There are initially 51 variables (50 proportions of RSRP from MR data and one GPS signal parameter) for the study area. As only two months of data are available for testing the contribution of the MR data for rainfall estimation, the driving parameters of the estimation model should be reduced through variable selection.

PCA is a multivariate statistical technique introduced by Pearson (1901) that transforms or orthogonally projects a larger series of correlated variables into a smaller set of uncorrelated variables, known as principal components (PCs), without losing too much information (Aitchison, 1983). PCA can be taken as a tool for reducing the high dimensionality of the original data while maintaining its variance. More details of the PCA method can be found in Jolliffe (2002). PCA has been widely used in various hydrological sciences and engineering problems, including defining the ecological hydrology of river (Chang et al., 2009), sustainability evaluation for biomass supply chain synthesis (How and Lam, 2018), and estimation of reservoir water saturation and water temperature (Asante-Okyere et al., 2020; Ren et al., 2020). Thus, in our study, PCA was used to reduce the initial 51 variables for wet-dry period classification and rainfall estimation.

3.2. Support vector classification (SVC) model for a wet-dry period classification

The main idea of the SVC model is to find a hyperplane in the data that is non-linearly transformed into a higher dimensional space through kernel methods (Suykens and Vandewalle, 1999). The optimal separating hyperplane between the two closest points on either side of the line is maximised to distinguish between classes by applying a kernel trick (Boser et al., 1992) and penalty term. The SVC model has been applied in wet-dry classification (Chen, et al., 2010) under climate change scenarios, in water quality assessment through data reduction (Singh et al., 2011; Tiyyasha, et al., 2020), in real-time water level prediction (Zhao et al., 2020), and so on. As there are two classes (i.e., wet (≥ 0.1 mm) and dry (< 0.1 mm)) in wet-dry period classification, the binary SVC algorithm was adopted here to improve rainfall estimation.

3.3. Artificial neural networks (ANN) method for rainfall estimation

ANN was implemented successfully in predicting and assessing hydrology (Hamed et al., 2004; Güclü and Dursun, 2008; Atasoy et al., 2013; Tiyyasha, et al., 2020) due to its robustness and capability for dealing with nonlinear and complex problems. ANN comprises a group of processing elements where each typical subgroup makes independent computations and passes their results to the next subgroup, which may perform in the same way. The network is often divided into three layers, i.e., subgroups of the processing elements. The three layers are the input layer, the hidden or intermediate layer, and the output layer. The input layer comprises the variables selected by the PCA method while the output layer outputs the rainfall simulation.

3.4. MISDC runoff hydrological model for runoff simulation

The “Modello Idrologico Semi-Distribuito in continuo” (MISDC) is a lumped hydrological model developed by Brocca et al. (2011). This conceptual model incorporates a two-layer continuous hydrological model, characterized by a component simulating the temporal pattern of soil moisture and a semi-distributed event-based rainfall-runoff component for simulating river discharge time series (Camici et al., 2020). By coupling these two components (models) through an experimentally derived relationship, the structure of the MISDC model was derived to simulate the rainfall-runoff transformation (Brocca et al., 2011) and the process of transferring the surface and subsurface runoff to the catchment outlet is simulated via the geomorphological

instantaneous unit hydrograph (Gupta et al. 1980). The MISDc model has been widely and successfully applied in many different contexts, including the mountainous catchment at an hourly time scale (Brocca et al., 2013; Barbetta et al., 2017; Loizu et al., 2018; Cislighi, et al., 2020). Thus, we consider the MISDc model suitable for the purpose of our study.

3.5. The performance criteria for the wet-dry period classification, rainfall estimation and runoff simulation

The classification error (CE) is often used for classification criteria and is employed to describe the performance of the SVC model.

$$CE = 1 - \frac{n}{N} \quad (1)$$

where n is the number of correct classifications and N is the total number of classifications.

The root mean squared error (RMSE) is used to measure the goodness of fit of the ANN method for rainfall estimation.

$$RMSE = \frac{1}{N} \sum_{i=1}^N (Q_{o,i} - Q_{s,i})^2 \quad (2)$$

where $Q_{o,i}$ is the i th observation runoff and $Q_{s,i}$ is the i th simulation runoff from the hydrological model. Smaller values of CN and $RMSE$ correspond to better model performance.

The Nash–Sutcliffe efficiency criterion (NSE, Nash and Sutcliffe, 1970), one of the most popular criteria used in hydrological modelling, is adopted here to quantitatively demonstrate the performance of the MISDc hydrological model.

$$NSE = 1 - \frac{\sum_{i=1}^N (Q_{o,i} - Q_{s,i})^2}{\sum_{i=1}^N (Q_{o,i} - \bar{Q}_{o,i})^2} \quad (3)$$

Larger values of NSE correspond to better model performance.

4. Results and discussion

4.1. The selected variables for the wet-dry period classification

Based on the two months (May and August 2019) of 30 min rainfall and MR data, there are a total of 2,593 records available for this study (some measurements are left out).

There are 50 proportions of RSRP from MR data as components in each record. If these components are input into the SVC model one by one for classifying the wet-dry period, the CE value vs the number of components can be determined. The 10-k cross-validation method was used to test the SVC model. All records were used as both training and testing samples and the results are shown in Figure 3(a). It was observed that the first eight components from PCA can cumulatively account for over 90 % of the total correct classification (i.e., the CE values are less than 0.1). Specifically, the average CE values are 0.028 and 0.08 for the training and testing samples, respectively. Thus, the first eight components from PCA based on the MR data are selected for the wet-dry period classification.

As the GPS signals can help evaluate the precipitable water vapour from the vertical path, both the GPS signals and the 50 proportions of RSRP from MR data are utilized for deriving components via the PCA method. Subsequently, all components were input into the SVC model with the 10-k cross-validation method and the corresponding results of average CE values are shown in Figure 3(b). It can be found that there are no significant differences between Figure 3(a) and Figure 3(b), indicating that the addition of GPS signals cannot significantly improve the performances of the wet-dry period classification. If there are only GPS signals for classifying wet-dry periods via the SVC model, the average CE values for both the training and testing samples are

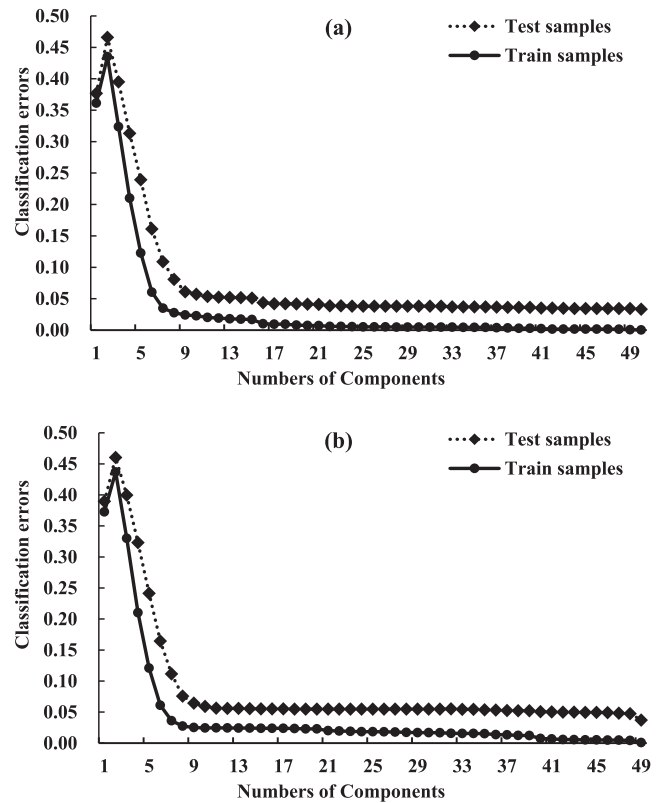


Figure 3. The average classification error (CE) values from the wet-dry period classification model (SVC) with 10-k cross validation method varying with the number of components from PCA method based on: (a) the proportions of RSRP from MR data; (b) both the GPS signals and the proportions of RSRP from MR data

approximately 0.31. The information contained in the GPS signals can achieve the most accurate classification of the wet-dry period. However, the first eight components of PCA from the proportions of RSRP were used to drive the SVC model for the wet-dry period classification due to their higher accuracy and relative concision.

4.2. Rainfall estimation via the ANN model

A total of 402 wet periods were found from the observed data. To determine the principal variables for rainfall estimation, 50 components (i.e., the variables derived from the MR data via the PCA method) were input into the rainfall estimation model (i.e., ANN that is the newff(●) function with {'tansig','purelin'} and 'trainrp' parameter values in Matlab). The 10-k cross-validation method was also used to determine the number of components for input variables and hidden nodes for the ANN parameter. The average RMSE values during both the calibration and validation periods vs the number of components are shown in Figure 4, when the RMSE was taken as the criteria for assessing the performance of the rainfall estimation model. It can be found that the first 18 PCA components cumulatively contribute the most to the rainfall estimation, where the change point of the RMSE values can be found. Therefore, these first 18 components, derived by the PCA method, were selected for driving the rainfall estimation ANN model. The number of nodes in the hidden layer determines the total number of parameters (i.e., the weights) of the ANN model to be calibrated and can impact the accuracy of rainfall simulation. The increased number of nodes in the hidden layer indicates an increase in the total parameters of the ANN model. To determine the number of hidden nodes in ANN, the average RMSE values in both the calibration and validation periods vs the number of hidden nodes in ANN were obtained through the 10-k cross-

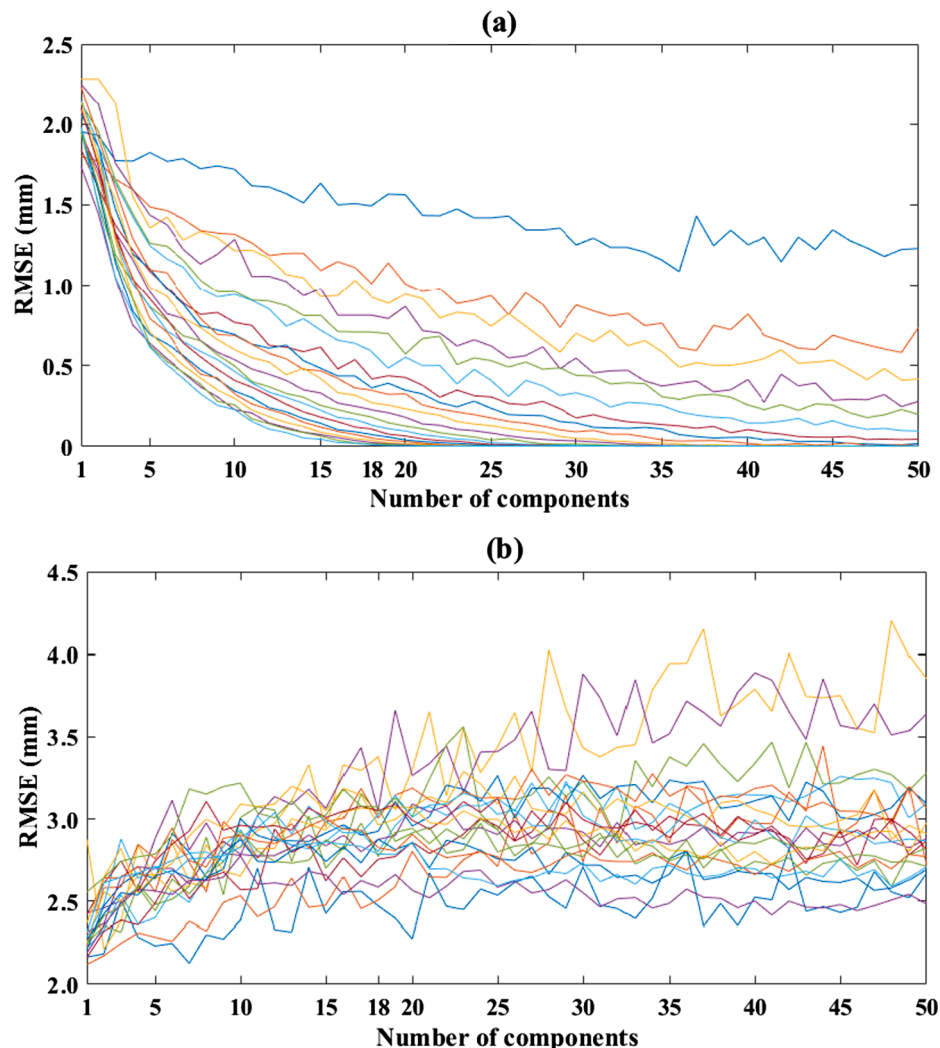


Figure 4. The RMSE values from the rainfall estimation (ANN model) varying with the number of components from PCA method based on the proportions of RSRP from MR data (Different curve represents the number of hidden nodes in ANN): (a) calibration; (b) validation.

validation method, as shown in Figure 5. It can be found that the seven hidden nodes in ANN can enable good performance for rainfall estimation with lower RMSE values during both the calibration and validation periods. To avoid overfitting of the ANN model limited by our available data sets, the number of nodes in the hidden layer was as seven. Therefore, the first eight components were selected as variables for the wet-dry period classification via the SVC model. The first 18 components from the PCA were taken as the input variables of the ANN model, with the structure of seven nodes in the hidden layer. When the classification results from the SVC were input into the rainfall estimation ANN model, as shown in Figure 2, the average RMSE values of rainfall estimation are 1.027 mm and 1.893 mm for the calibration and validation periods, respectively, using the 10-k cross-validation method.

To determine the applicability of the rainfall retrieval process during runoff simulations, the first 60 % of the wet and dry records were taken as training samples for classification while the remaining 40 % were taken as testing samples. The *EC* values are 0.018 and 0.102 for the training and testing samples, respectively. There were 402 wet periods from the observed data and 480 wet periods can be found from the classification results by the SVC model. It is interesting to find that all 402 wet periods could be determined while 78 wet periods were misclassified by the SVC model. 63 % of wet periods occurred between 4:00 and 16:00; however, 50 % of misclassification can be found during this period. The highest number of misclassifications occurred at

approximately 10:00 and 22:00. (shown in Figure 6.); this might be attributed to the peak times of cell phone use. The lowest occurred from 12:00 to 14:00 (i.e., lunchtime). According to the rainfall retrieval process, rainfall estimation should be conducted based on the results of wet-dry period classification. The first 60 % of wet records from the SVC classification model were taken as training samples for ANN and the remaining 40 % were taken as testing samples. The results of rainfall estimation from the ANN model are shown in Figure 7 (i.e., the `newff()` function with “[7,1], {‘tansig’, ‘purelin’}, ‘trainrp’ ” parameters values in Matlab). The RMSE values are 0.911 mm and 0.931 mm for the training (calibration) and testing (validation) periods, respectively. The rainfall estimation error is mainly attributed to misclassifications from the SVC model, as shown in Figure 7. The accuracy of wet period classification is important for improving the accuracy of rainfall estimation. Although large rainfall values appear to be underestimated due to few large value records for training and testing (as shown in Figure 7(a) and 7 (b)), the accuracies of the ANN model driven by the main components from MR data are acceptable.

4.3. The effects of the estimated rainfall from MR data on runoff simulation

A predominant application of rainfall monitoring/estimation is in hydrological models for runoff process simulations to aid flood

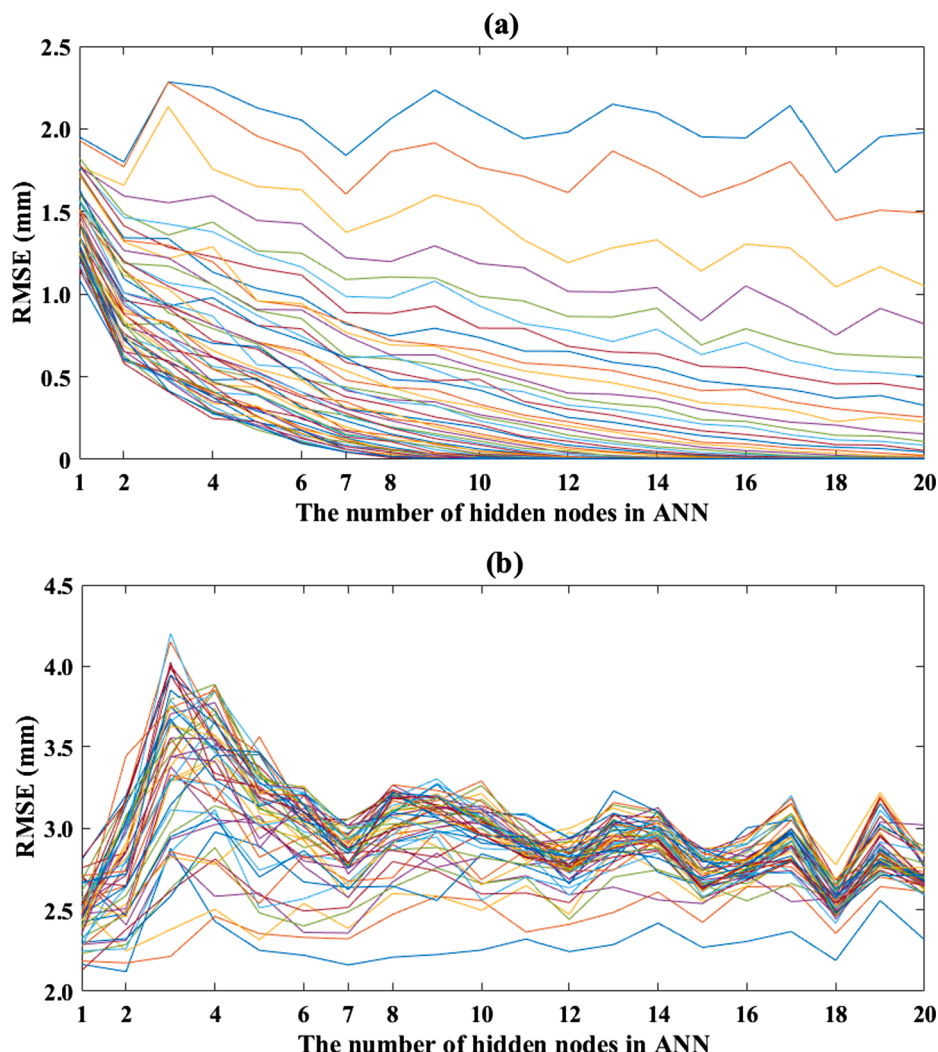


Figure 5. The effects of the number of hidden nodes in ANN on the RMSE values of rainfall estimation during the wet period (Different curve represents the number of components): (a) calibration; (b) validation

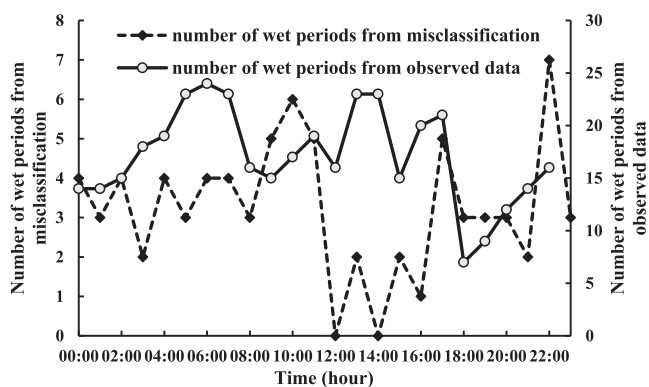


Figure 6. Number of wet periods from the observed data and from the misclassification by SVC model.

forecasting or water resources management and, thus, the estimated rainfall by ANN model was input into the MISDC runoff hydrological model to evaluate its effects. According to the variation in the discharge series during May and August 2019, for the intended study area, $44 \text{ m}^3/\text{s}$ (i.e., the average discharge during the observed period) was set as the threshold for extracting the flood events; 39 flood events were found

during the two months. The lumped MISDC runoff hydrological model was calibrated via the observed rainfall (areal rainfall) and runoff data from the gauge stations. Then, the runoff simulation could be obtained as shown in Figure 8(a₁), with an NSE value of 0.967. The scatter plot between the observed and simulated runoff is shown in Figure 8(a₂). The calibrated MISDC runoff hydrological model can simulate runoff with acceptable accuracy (>0.8); even the magnitudes of runoff exceed $370 \text{ m}^3/\text{s}$. However, most runoff between $200 \text{ m}^3/\text{s}$ and $370 \text{ m}^3/\text{s}$ is underestimated by the MISDC model while runoff between $50 \text{ m}^3/\text{s}$ and $200 \text{ m}^3/\text{s}$ is slightly overestimated. If the estimated rainfall is input into the calibrated MISDC model, its runoff simulation can also be obtained, as shown in Figure 8 (b₁) and (b₂). The NSE value is 0.934 and is lower than that from the observed data shown in Figure 8(a₁). Taking the runoff simulation from the observed rainfall data as a reference, the differences between these simulations can reflect the effects of the estimated rainfall from MR data on runoff simulation (shown in Figure 8 (b₂)). The details of the rainfall retrieval process can be found in the last paragraph in Section 4.2. Some rainfall events have been significantly underestimated by the ANN model (lower than line 1:1, shown in Figure 7) and their corresponding runoffs are also underestimated to the observed runoff (shown in Figure 8 (b₂)), especially the runoff from 120 to $350 \text{ m}^3/\text{s}$. All wet periods were incorporated by the SVC classification for the rainfall estimation ANN model and lower rainfall estimation corresponded to lower runoff simulation. It is found that most runoff

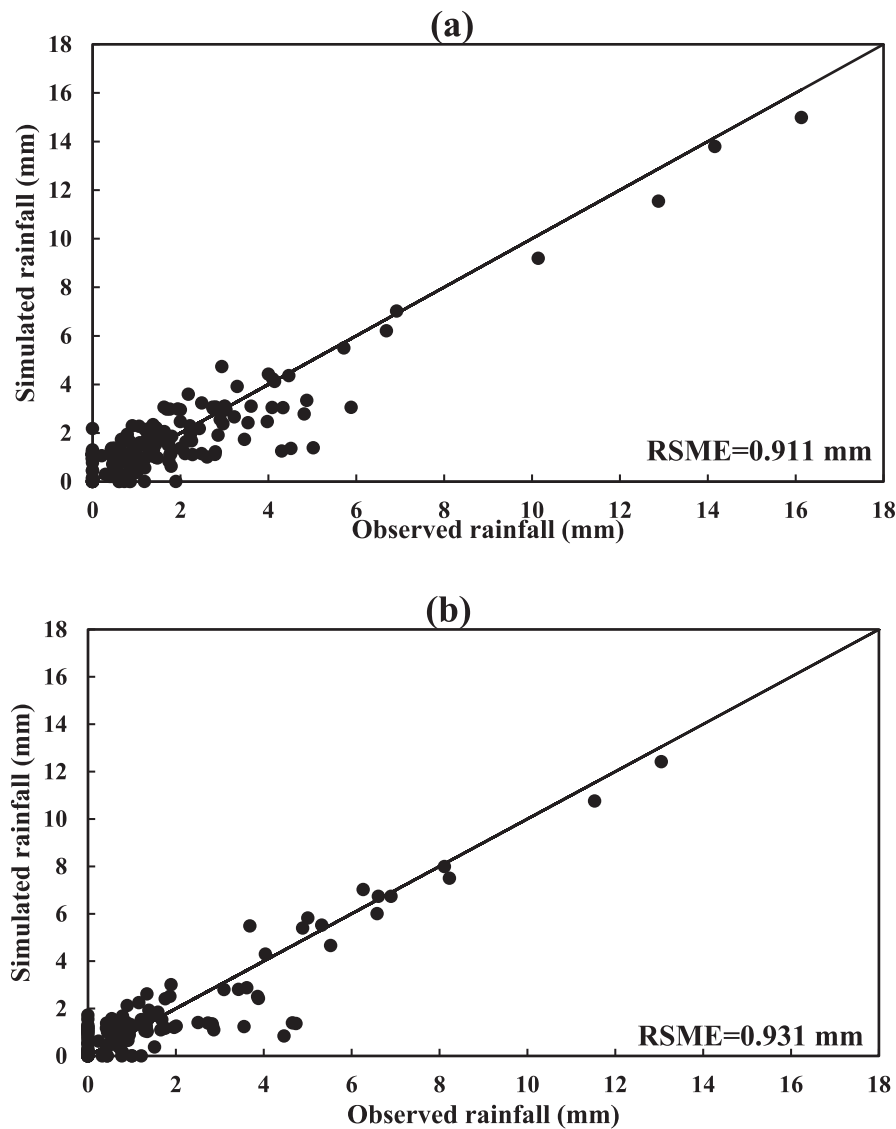


Figure 7. The rainfall estimation from ANN model based on the classification results from the SVC model: (a) training (calibration); (b) testing (validation).

estimation is lower from the estimated rainfall than from observed rainfall; this can be attributed to the effect of rainfall estimation on the runoff simulation.

5. Conclusion

MR data from the TD-LTE MR data acquisition system can potentially be used for rainfall estimation. Real time MR data provided by the communication company offers great opportunities for monitoring rainfall at high temporal and spatial scales, without additional cost. The purpose of MR data is to reflect the coverage and interference of the TD-LTE network (i.e., the services) for the customer and is an indicator of the microwave link between the telephone tower and cell phone or mobile terminal. Differing from current rainfall retrieval algorithms based on the radio connection between telephone towers, the methods of estimating rainfall in our study based on the MR data will not be replaced by the development of underground fibre-optic cable networks. The MR data is just the proportion of weak coverage samples and can still be used to estimate rainfall owing to the distribution of proportions directly impacted by the rainfall-based attenuation.

MR data during May and August, 2019, from the Huizhou branch of China Mobile Communications Group Guangdong, were converted to

rainfall through our proposed framework and models, which were calibrated via local observed rainfall and runoff data. The promising results of the estimated rainfall and its corresponding simulated runoff have proven that the MR data and our proposed models can provide a new method of monitoring rainfall and simulating runoff.

There are still several improvements to be conducted in future studies. The major error in rainfall estimation occurs for wet-dry period classification. Another wet-dry classification algorithm or merging of different rainfall sources (e.g., satellite rainfall data, if available on time) might improve the accuracy of wet-dry period classification. The structure of the rainfall estimation ANN model might be further optimized via dropout to prevent neural networks from overfitting. Long MR data sets from different places in different seasons should be adopted to further evaluate the accuracy of rainfall estimation and to determine the factors that impact the effectiveness of the proposed method.

CRediT authorship contribution statement

Dedi Liu: Conceptualization, Methodology, Software, Data curation, Writing - review & editing. **Yurong Zhang:** . **Jianming Zhang:** . **Lihua Xiong:** Investigation. **Pan Liu:** Investigation. **Hua Chen:** Investigation. **Jiabo Yin:** Investigation.

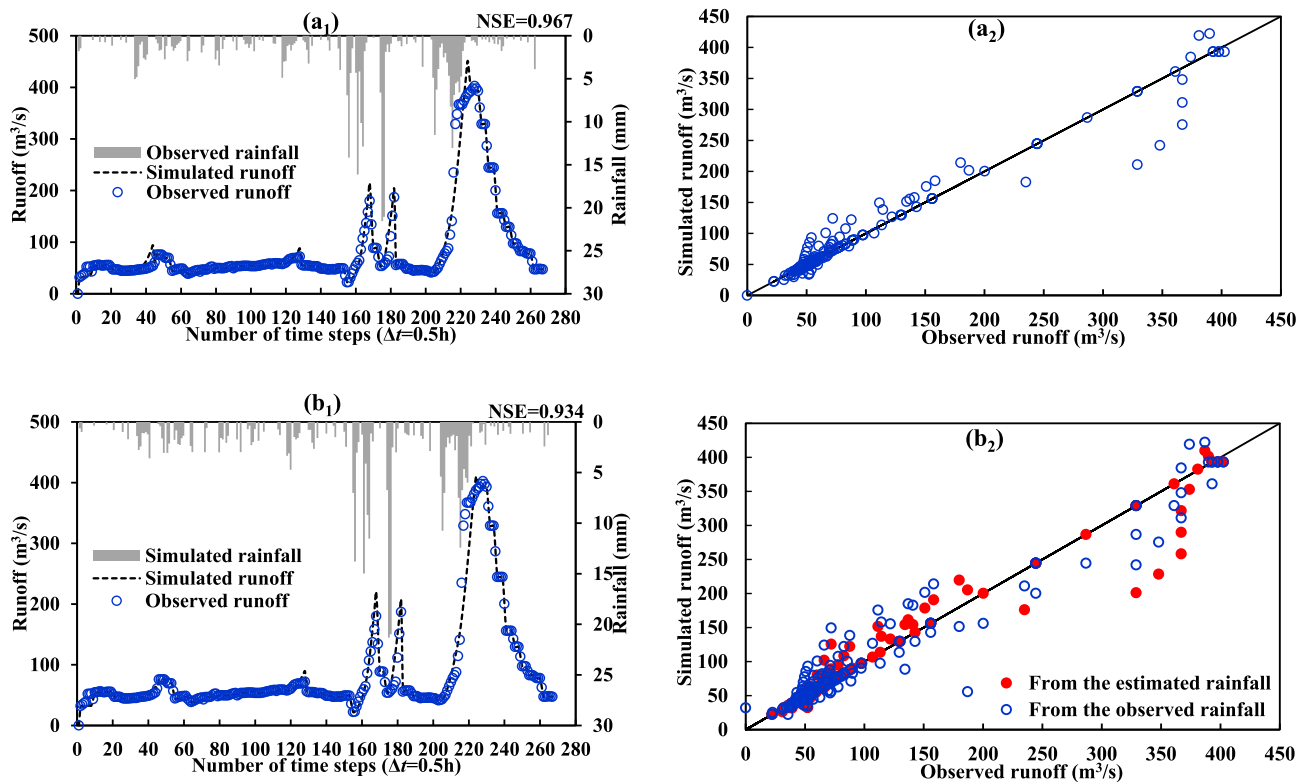


Figure 8. Runoff simulation from the MISDC model calibrated by the observed rainfall data: (a₁) runoff simulation driving by the observed gauge rainfall; (a₂) scatter plot of the runoff observation and simulation from the observed rainfall data; (b₁) runoff simulation driving by the estimated rainfall from MR data; (b₂) scatter plot of the runoff observation and simulation from the observed rainfall data and the estimated rainfall from MR data

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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